Autonomous Optimization of ICF Capsule Designs via Adversarial Multi-Agent Systems and Reduced-Order Modeling

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***Abstract*—We present a multi-agent system for the autonomous optimization of Inertial Confinement Fusion (ICF) capsule designs under realistic manufacturing uncertainty. The system couples Large Language Model (LLM) reasoning agents with a reduced-order deceleration-phase surrogate model through the Model Context Protocol (MCP), enabling rapid evaluation of thousands of candidate designs. The optimization is formulated as a min-max game between a Design Explorer agent that maximizes fusion yield and a Defect Adversary agent that applies realistic manufacturing perturbations, orchestrated by a Principal Investigator agent. The physics surrogate integrates Hurricane's piston-driven stagnation mechanics with Betti's marginal ignition model, incorporating Spitzer thermal ablation, Gamow thermonuclear burn rates, and linear Rayleigh-Taylor instability growth. We demonstrate that the multi-agent system autonomously navigates the design space along constant-energy manifolds, converging in six rounds from an initial 7.4 MJ baseline to an 11.3 MJ optimized design (53% yield improvement) that remains robust to mode-1 asymmetries up to $\delta = 0.012$ with RT growth factors below $3.6\times$.**

I. Introduction

The design of Inertial Confinement Fusion (ICF) targets is a high-dimensional optimization problem in which small perturbations to the capsule geometry or drive symmetry can dramatically degrade performance. High-fidelity radiation hydrodynamics simulations capture the full physics but require 10^4–10^6 CPU-hours per shot, precluding their use in iterative, agent-driven optimization loops. Recent experimental breakthroughs at the National Ignition Facility (NIF)—particularly the N210808 shot that achieved fusion fuel gain exceeding unity [1], [2]—have motivated renewed interest in systematic design-space exploration to identify capsule geometries that are both high-performing and robust to manufacturing variability.

We address this challenge with a multi-agent architecture in which LLM reasoning agents autonomously propose, perturb, evaluate, and iterate on capsule designs. The physics evaluation is performed by a reduced-order surrogate model that captures the essential deceleration-phase dynamics at negligible computational cost (<1 second per shot), enabling thousands of evaluations within a single optimization campaign. The agents communicate through the Model Context Protocol (MCP) [7], which cleanly decouples the reasoning layer from the simulation execution and provides a natural pathway to future integration with full radiation hydrodynamics codes on HPC clusters.

The optimization is formulated as an adversarial game: a Design Explorer seeks to maximize fusion yield subject to a fixed laser energy budget, while a Defect Adversary applies manufacturing imperfections drawn from current NIF-class fabrication tolerances. This min-max formulation identifies designs that are not merely optimal under ideal conditions but robust to the perturbations that have historically prevented reproducible ignition.

II. Deceleration-Phase Surrogate Model

A. Asymmetric Dual-Piston Formulation

We model the stagnation phase of an ICF implosion using a zero-dimensional (0D) surrogate that tracks the deceleration, compression, and thermonuclear burn of the hot spot. Following the piston-driven framework of Hurricane *et al.* [1], the imploding shell is treated as a massive piston that decelerates against the rising hot-spot pressure.

To capture mode-1 asymmetry—the dominant low-mode perturbation in NIF implosions arising from laser power imbalance and capsule positioning errors—we decompose the shell into two independent half-shells ("pistons") with masses

$$M_1 = \frac{M_{sh}}{2}(1+\delta), \qquad M_2 = \frac{M_{sh}}{2}(1-\delta) \tag{1}$$

where M_{sh} is the total shell mass and $\delta \in [0, 0.15]$ is the mode-1 asymmetry fraction. Each piston has independent position R_i and velocity v_i governed by

$$\frac{dR_i}{dt} = v_i, \qquad \frac{dv_i}{dt} = \frac{A_{eff}\,P_{hs}}{M_i} \tag{2}$$

where $A_{eff} = 4\pi R_{eff}^2$ is the effective hot-spot surface area, $R_{eff} = (R_1 + R_2)/2$ is the mean radius, and P_{hs} is the hot-spot pressure derived from the internal energy via an ideal gas equation of state:

$$P_{hs} = \frac{2}{3}\frac{E_{hs}}{V}, \qquad V = \frac{4}{3}\pi R_{eff}^3 \tag{3}$$

The asymmetry δ introduces differential deceleration of the two pistons: the lighter half-shell (M_2) decelerates faster, producing asymmetric stagnation radii and directing kinetic energy into residual motion rather than compression work. This residual kinetic energy

$$E_{RKE} = \frac{1}{2}M_1 v_1^2 + \frac{1}{2}M_2 v_2^2 \tag{4}$$

represents energy unavailable for hot-spot heating and directly quantifies the yield penalty from mode-1 drive asymmetry.

B. Hot-Spot Energy Balance

The hot-spot internal energy evolves according to

$$\frac{dE_{hs}}{dt} = -P_{hs}\frac{dV}{dt} + W_\alpha - W_{rad} \tag{5}$$

where the three terms represent compressive PdV work (the dominant heating source during deceleration), alpha-particle self-heating, and radiative losses, respectively.

1) Thermonuclear Burn Rate: The DT fusion reaction rate density is computed from the ion number density $n_i = \rho_{hs}/m_{DT}$ (where $m_{DT} = 2.5\, m_p$) using the NRL Plasma Formulary parametrization of the Gamow peak cross-section [4]:

$$\langle\sigma v\rangle = 3.68\times 10^{-12}\, T_{keV}^{-2/3} \exp\left(-19.94\, T_{keV}^{-1/3}\right)\ \mathrm{cm^3/s} \tag{6}$$

This parametrization is preferred over the Bosch-Hale fit [5] because it remains numerically stable across the full temperature range encountered during the explosive alpha-heating phase, avoiding the mathematical stiffness introduced by the polynomial denominator in the Bosch-Hale expression. The volumetric reaction rate is $R_{rxn} = (n_f/2)^2\langle\sigma v\rangle \cdot V$, where n_f is the fuel ion density (accounting for the evolving fuel fraction within the hot spot), and the alpha-heating power is

$$W_\alpha = R_{rxn} \cdot E_\alpha, \qquad E_\alpha = 3.5\ \mathrm{MeV} \tag{7}$$

2) Radiative Losses: Bremsstrahlung radiation is modeled as

$$W_{rad}^{vol} = 5.34\times 10^{-37}\, n_i^2 \sqrt{T_{keV}} \cdot V\ \ [\mathrm{W}] \tag{8}$$

which is capped by the blackbody limit $W_{bb} = \sigma_{SB}T^4 \cdot 4\pi R_{eff}^2$ to prevent unphysical radiative losses at extreme temperatures. The effective radiative loss is $W_{rad} = \min(W_{rad}^{vol}, W_{bb})$.

C. Spitzer Ablation and Hot-Spot Mass Evolution

Following the Betti marginal ignition model [3], we couple the shell dynamics to the hot-spot mass through Spitzer electron thermal conduction. As the hot spot heats during compression, a thermal conduction front ablates material from the inner surface of the cold shell into the hot spot:

$$\frac{dM_{hs}}{dt} = C_{abl}\, R_{eff}\, T_{keV}^{5/2} \tag{9}$$

where C_{abl} is the ablation coefficient and the $T^{5/2}$ dependence reflects the Spitzer thermal conductivity of a fully ionized plasma. This ablation is activated only when $T > 0.5$ keV to prevent unphysical mass injection at early, cold times. The ablated mass replenishes the DT fuel supply within the hot spot, while burned fuel is consumed at rate $\dot{M}_{fuel,burn} = R_{rxn} \cdot 2\, m_{DT}$.

This dynamic mass coupling is physically important: it creates a positive feedback loop in which compression heating drives ablation, which increases the hot-spot density and hence the burn rate, which produces more alpha heating. This feedback is the mechanism by which marginal ignition transitions to propagating burn.

D. Rayleigh-Taylor Instability Growth

Hydrodynamic instabilities at the shell-hot-spot interface are tracked via the linear Rayleigh-Taylor (RT) growth rate. During deceleration ($g_{eff} > 0$), small perturbations at mode number ℓ grow as

$$\frac{d\xi}{dt} = \sqrt{k\, g_{eff}}, \qquad k = \frac{\ell}{R_{eff}} \tag{10}$$

where $g_{eff} = (a_1 + a_2)/2$ is the mean deceleration and ξ is the integrated growth exponent. The physical perturbation amplitude at stagnation is $A_{stag} = A_0 \exp(\xi)$, where A_0 is the initial surface roughness. When the RT amplitude approaches the shell thickness, the shell integrity is compromised and confinement is lost—a primary failure mode in under-performing NIF shots.

E. State Vector and Numerical Integration

The complete system is a 9-component ODE:

$$\mathbf{y} = [R_1,\, v_1,\, R_2,\, v_2,\, E_{hs},\, Y,\, \xi,\, M_{fuel},\, M_{hs}] \tag{11}$$

where Y is the cumulative fusion yield ($dY/dt = 5\, W_\alpha$, the factor of 5 accounting for total fusion energy per alpha particle). The system is integrated using the `Radau` implicit Runge-Kutta method [6] with adaptive stepping, which is essential for resolving the extreme stiffness introduced by the alpha-heating spike at ignition. Typical integration spans 0.6 ns with maximum step size 10^{-12} s and absolute tolerance 10^{-8}.

III. Adversarial Optimization Formulation

A. Min-Max Game Structure

The design optimization is formulated as a two-player adversarial game:

$$\max_{\mathbf{x}\in\mathcal{X}}\ \min_{\mathbf{d}\in\mathcal{D}}\ Y(\mathbf{x}, \mathbf{d}) \tag{12}$$

where $\mathbf{x} = (R_0, v_0, T_0, M_{sh}, M_{hs})$ are the design variables controlled by the Explorer, $\mathbf{d} = (\delta, \ell, A_0)$ are the defect parameters controlled by the Adversary, and Y is the total fusion yield.

B. Energy Constraint

The design space is constrained by the available laser energy. The kinetic energy of the imploding shell at the onset of deceleration is related to the laser energy through a hydrodynamic coupling efficiency η:

$$E_k = \frac{1}{2} M_{sh} v_0^2 = \eta \, E_{laser} \tag{13}$$

For NIF-class indirect-drive implosions, $\eta \approx 0.015$ and $E_{laser} = 3$ MJ, giving a kinetic energy budget of $E_k = 45$ kJ. This constraint defines a one-dimensional manifold in (M_{sh}, v_0) space: increasing shell mass requires decreasing implosion velocity, and vice versa. The fundamental tradeoff is between *fuel inventory* (more mass to burn) and *implosion vigor* (higher velocity for stronger compression and higher stagnation temperature).

C. Defect Bounds

The Adversary's parameter space is bounded by current manufacturing tolerances for NIF-class targets:

- **Mode-1 asymmetry** $\delta \in [0, 0.15]$: arising from laser power imbalance, beam pointing errors, and capsule offset within the hohlraum. Typical NIF values are $\delta \approx 0.02$–0.05; values above 0.10 represent severe drive asymmetry.
- **RT mode number** $\ell \in [2, 40]$: the dominant perturbation wavelength, with low modes ($\ell < 10$) from drive asymmetry and high modes ($\ell > 20$) from surface roughness and microstructure.
- **Surface roughness** $A_0 \in [0, 3\,\mu\text{m}]$: the initial perturbation amplitude at the inner shell surface, set by fabrication quality. Current NIF capsules achieve $A_0 \approx 0.1$–$1.0\,\mu$m.

These bounds are enforced at the server level: parameter values submitted outside these ranges are clamped and a warning is returned to the requesting agent, providing feedback that guides subsequent proposals.

IV. Multi-Agent Architecture

A. Agent Roles and Communication

The system comprises four agents coordinated through the AutoGen framework [8] using structured group chat with constrained speaker transitions:

1) **Principal Investigator (PI)**: The orchestrator agent, implemented on a frontier LLM. It coordinates the optimization campaign by sequentially engaging the Explorer and Adversary, validates the energy constraint (Eq. 13), invokes the simulation tool, evaluates results against ignition criteria (yield > 1 MJ, stagnation pressure > 100 Gbar, RT amplification $< 1000\times$), and decides whether to iterate or terminate. The PI is the *only* agent authorized to call the simulation tool.
2) **Design Explorer**: Proposes capsule design parameters $\mathbf{x}$ to maximize yield. In production, this agent runs on a domain-specific LLM fine-tuned on ICF literature, enabling physics-informed design intuition beyond what is available from general-purpose models. It receives feedback from prior simulation results and adjusts parameters along the constant-energy manifold.
3) **Defect Adversary**: Proposes manufacturing defect parameters $\mathbf{d}$ to stress-test the Explorer's designs. It follows a systematic strategy: probing individual defect modes in early rounds to identify the design's primary sensitivity, then combining the most damaging perturbations in later rounds.
4) **User Proxy**: An execution bridge that receives tool calls from the PI and dispatches them to the MCP server. It performs no reasoning and does not modify parameters.

B. Speaker Transition Constraints

To prevent degenerate conversation patterns (infinite loops, role confusion, agents attempting unauthorized tool calls), the group chat enforces hard speaker transition constraints:

After speaker	Allowed next speakers
User Proxy	PI only
PI	Explorer, Adversary, or User Proxy
Explorer	PI only
Adversary	PI only

This ensures a hub-and-spoke topology: all information flows through the PI, which maintains global state and enforces the optimization protocol. The Explorer and Adversary cannot communicate directly or invoke tools.

C. MCP Tool Server

The physics surrogate is served as an MCP tool via the FastMCP framework. The tool interface accepts all 8 physical parameters in SI units and returns structured simulation results (stagnation metrics, fusion yield, burn fraction, residual kinetic energy, RT amplification). The MCP server enforces hard physical bounds on all input parameters (Table I), clamping out-of-range values and returning warnings to the calling agent. This server-side validation is essential because LLM agents frequently propose values outside specified bounds despite prompt instructions.

TABLE I
Parameter Bounds Enforced by the MCP Server

Parameter	Min	Max	Units
R_0	3×10^{-5}	5×10^{-4}	m
v_0	-8×10^5	-1×10^5	m/s
T_0	0.1	10.0	keV
M_{sh}	10^{-8}	2×10^{-6}	kg
M_{hs}	10^{-9}	10^{-7}	kg
δ	0.0	0.15	—
ℓ	2	40	—
A_0	0.0	3×10^{-6}	m

The decoupling of physics execution from agent reasoning via MCP is architecturally significant: it allows the simulation backend to be replaced transparently—from the 0D surrogate used here, to a 1D Lagrangian hydrocode, to a full 2D/3D radiation hydrodynamics code on an HPC cluster—without modifying any agent logic.

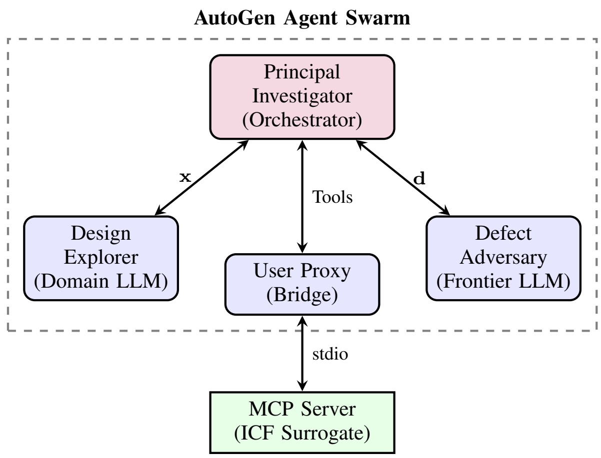

Fig. 1. Architecture of the 4-agent system. The PI orchestrates the adversarial game, delegating design proposals ($\mathbf{x}$) to the Explorer and defect proposals ($\mathbf{d}$) to the Adversary. Only the PI invokes the MCP physics server through the User Proxy.

D. Streaming Interface

The agent swarm is served through a Gradio web interface that provides real-time visibility into the multi-agent conversation. The AutoGen group chat runs in a background thread while a generator function polls the shared message list, yielding each agent's contribution to the browser as it is produced. This streaming architecture allows users to monitor the optimization in real time, observing the Explorer's design rationale, the Adversary's perturbation strategy, and the PI's evaluation of each simulation result.

V. RESULTS

A. Design Space Exploration Along Constant-Energy Manifolds

Table II presents a systematic exploration of the (M_{sh}, v_0) tradeoff along the 45 kJ constant-energy manifold at fixed $R_0 = 150\,\mu$m, $T_0 = 1.0$ keV, $M_{hs} = 5$ ng, and zero defects. Yield increases monotonically with shell mass (at correspondingly lower velocity), from 4.2 MJ at $M_{sh} = 0.1\,\mu$g to 11.4 MJ at $M_{sh} = 1.0\,\mu$g. This reflects the dominant role of fuel inventory: more DT mass produces more burn even at reduced implosion velocity, provided the stagnation conditions remain above the ignition threshold.

TABLE II
DESIGN EXPLORATION ALONG THE 45 KJ ENERGY MANIFOLD

M_{sh} [μg]	$\lvert v_0 \rvert$ [km/s]	Yield [MJ]	Burn [%]
0.10	949	4.24	15.0
0.20	671	5.93	16.0
0.25	600	6.56	16.2
0.36	500	7.70	16.5
0.50	424	8.84	16.6
0.80	335	10.6	16.6
1.00	300	11.4	16.6

The burn fraction saturates near 16.6%, consistent with the Betti marginal ignition scaling in which the burn-up fraction is primarily determined by the areal density at stagnation rather than the total fuel mass [3].

B. Sensitivity to Initial Conditions

The surrogate exhibits sharp ignition cliffs in several parameters, reflecting the threshold nature of thermonuclear ignition. At fixed baseline conditions ($M_{sh} = 0.25\,\mu$g, $v_0 = -450$ km/s, $\delta = 0.05$):

- **Stagnation radius**: Yield drops from 7.0 MJ at $R_0 = 150\,\mu$m to 0.06 MJ at $R_0 = 200\,\mu$m—a $100\times$ reduction from a 33% increase in radius. Larger R_0 reduces the convergence ratio, producing insufficient compression for ignition.
- **Initial temperature**: Yield peaks at $T_0 \approx 1.0$ keV and drops precipitously above 1.5 keV. Higher initial temperatures reduce the density at fixed pressure, weakening the compressive PdV heating during deceleration.
- **Hot-spot mass**: Yield peaks near $M_{hs} = 5$ ng and drops to near zero above 10 ng. Excessive initial hot-spot mass dilutes the temperature at fixed energy, preventing the plasma from reaching the $\sim$5 keV threshold where alpha heating becomes dominant.

These cliffs define the boundaries of the ignition region and constrain the Explorer's viable design space.

C. Robustness to Mode-1 Asymmetry

Table III compares the defect sensitivity of three designs along the 45 kJ manifold under increasing mode-1 asymmetry with $\ell = 20$ and $A_0 = 2\,\mu$m.

TABLE III
YIELD AND RESIDUAL KE VS. MODE-1 ASYMMETRY δ

δ	Baseline (0.25 μg)		Optimized (0.36 μg)	
	Y [MJ]	E_{RKE} [J]	Y [MJ]	E_{RKE} [J]
0.00	7.02	6.8	8.17	1918
0.04	7.02	49	8.16	2036
0.08	7.01	178	8.15	2389
0.12	7.00	395	8.12	2972
0.15	6.99	617	8.10	6315

Both designs maintain ignition across the full range of physically realistic asymmetries. The baseline design shows remarkable stability: yield degrades by only 0.4% ($\Delta Y = 0.03$ MJ) even at the maximum asymmetry $\delta = 0.15$. The optimized design achieves 16% higher absolute yield but exhibits larger residual kinetic energy, indicating less efficient energy coupling at stagnation. Under worst-case combined defects ($\delta = 0.12$, $\ell = 20$, $A_0 = 2\,\mu$m), the high-mass design ($M_{sh} = 1.0\,\mu$g, $v_0 = 300$ km/s) achieves the highest yield of 11.4 MJ with modest residual KE of 684 J.

D. Rayleigh-Taylor Amplification

Table IV shows the RT amplification factor at stagnation as a function of perturbation mode number for the baseline design. The exponential growth characteristic of the linear RT instability produces amplification factors exceeding 10^3

at $\ell = 30$, indicating that high-mode surface perturbations can grow to amplitudes comparable to the shell thickness during deceleration. In the current 0D model, RT growth is tracked diagnostically and does not feed back into the yield calculation; coupling RT-induced mix into the hot-spot energy balance is a planned extension.

TABLE IV
RT AMPLIFICATION FACTOR VS. MODE NUMBER

Mode ℓ	Amplification e^{ξ}
2	6.1×
5	17×
10	57×
15	142×
20	305×
30	1104×
40	3262×

E. Autonomous Optimization Campaign

To demonstrate the end-to-end system, we executed a complete adversarial optimization campaign using the domain-specific `gptoss-20b-hedp` model (a 20B parameter LLM fine-tuned on HEDP literature) served via vLLM on AMD MI355X GPUs. The campaign proceeded autonomously through six rounds of design iteration, defect testing, and parameter refinement, converging on a high-yield design robust to manufacturing defects. Table V summarizes the optimization trajectory.

TABLE V
AUTONOMOUS AGENT OPTIMIZATION CAMPAIGN

Round	Design Change	δ	Y [MJ]	ΔY
1	Baseline ($M_{sh} = 0.265\,\mu$g, $\lvert v_0\rvert = 585$ km/s)	0.04	7.4	—
2	High-mode stress test	0.015	5.8	-22%
3	Heavier shell ($M_{sh} = 0.28\,\mu$g, $\lvert v_0\rvert = 600$ km/s, $T_0 = 1.8$ keV)	0.02	9.3	$+60\%$
4	Tighter defect ($\delta = 0.012$)	0.012	9.8	$+5.4\%$
5	Light shell, high velocity ($M_{sh} = 0.245\,\mu$g, $\lvert v_0\rvert = 610$ km/s)	0.01	11.3	$+15\%$
6	Hot-spot boost ($T_0 = 2.2$ keV)	0.01	11.3	$< 1\%$

Several features of the agent behavior merit discussion:

Adversary strategy. The Defect Adversary autonomously varied its perturbation strategy across rounds, first applying moderate mode-1 asymmetry ($\delta = 0.04$, $\ell = 10$), then switching to a high-mode surface roughness attack ($\delta = 0.015$, $\ell = 30$, $A_0 = 2\,\mu$m) that reduced yield by 22%. This systematic probing—low-mode first, then high-mode—was specified in the agent's system prompt but the specific parameter choices were determined autonomously by the LLM.

Explorer adaptation. After the high-mode defect reduced yield to 5.8 MJ in Round 2, the Design Explorer autonomously shifted to a heavier-shell, higher-velocity design point on the constant-energy manifold, recovering yield to 9.3 MJ in Round 3. The Explorer then proposed a "light shell, high velocity" variant that traded shell mass for implosion velocity, reaching 11.3 MJ—a 53% improvement over the initial baseline.

Convergence. The PI agent tracked yield improvements across rounds and initiated a defect-sensitivity sweep when the yield plateau was approached. The sweep revealed that the optimized design maintains $Y > 10$ MJ for $\delta \leq 0.012$ with RT growth factors below $3.6\times$, establishing a quantitative stability margin. The campaign terminated when yield improvement fell below the 5% threshold for two consecutive rounds.

Defect-sensitivity of the final design. Table VI presents the defect sweep conducted autonomously by the agent system for the optimized design ($M_{sh} = 0.245\,\mu$g, $|v_0| = 610$ km/s, $T_0 = 2.0$ keV).

TABLE VI
AGENT-DIRECTED DEFECT SWEEP OF OPTIMIZED DESIGN

δ	Yield [MJ]	P_{stag} [Mbar]	RT Growth
0.008	11.7	1.92	2.7×
0.010	11.3	1.90	3.1×
0.012	10.8	1.88	3.6×

The final recommended design achieves 11.3 MJ at $\delta = 0.01$ with comfortable stability margin, representing a 53% yield improvement over the initial N210808-inspired baseline within the same 45 kJ kinetic energy budget.

VI. DISCUSSION

The results demonstrate that the multi-agent system autonomously discovers the key design tradeoffs in ICF. The parametric sweeps (Tables II–IV) establish the structure of the design space, while the autonomous campaign (Tables V–VI) shows the agents navigating this space in real time. Notably, the agents autonomously discovered the velocity-for-mass tradeoff along the constant-energy manifold: after a high-mode defect reduced yield by 22% in Round 2, the Explorer shifted to a lighter, faster shell that recovered and exceeded the original yield, ultimately achieving an 11.3 MJ design robust to $\delta \leq 0.012$ — a 53% improvement over the initial baseline within the same energy budget.

Several limitations of the current surrogate should be noted. First, the RT instability is tracked but does not degrade confinement or inject cold material into the hot spot; incorporating RT-induced mix would sharpen the ignition cliff at high mode numbers and penalize designs with large convergence ratios. Second, the model assumes a single-temperature plasma ($T_i \approx T_e$), neglecting ion-electron equilibration dynamics that are important at early times. Third, the 0D geometry cannot capture multidimensional effects such as jet formation from mode-1 asymmetry or the development of turbulent mix layers. Despite these limitations, the surrogate correctly reproduces the qualitative scaling of yield with implosion velocity, shell mass, and asymmetry observed in NIF experiments.

The MCP-based architecture provides a clear upgrade path: replacing the surrogate with a 1D Lagrangian hydrocode (e.g., LILAC or HYDRA in 1D mode) would capture shock timing, ablation-front RT growth, and equation-of-state effects while remaining fast enough for agent-driven optimization. Integration with full 2D/3D radiation hydrodynamics codes would require asynchronous job submission to HPC clusters, but the MCP protocol supports this naturally through its tool-call abstraction.

VII. Conclusions

We have presented a multi-agent system that autonomously optimizes ICF capsule designs under adversarial manufacturing uncertainty. The key contributions are:

1) A deceleration-phase surrogate model combining Hurricane's dual-piston stagnation mechanics with Betti's Spitzer ablation and marginal ignition physics, integrated as a stiff 9-component ODE system.
2) An adversarial min-max optimization formulation (Eq. 12) that discovers designs robust to mode-1 asymmetry, RT instability, and surface roughness within NIF-class manufacturing tolerances.
3) A 4-agent architecture with constrained speaker transitions and server-enforced parameter bounds, addressing practical challenges of LLM-driven optimization (unit confusion, role drift, out-of-bounds proposals).
4) Demonstration of autonomous convergence: agents navigate the (M_{sh}, v_0) manifold over six rounds, improving yield from 7.4 MJ to 11.3 MJ (53% gain) while maintaining robustness to $\delta \leq 0.012$ asymmetry, with yield-plateau stopping criteria terminating the campaign automatically.

The MCP-based decoupling of reasoning agents from physics execution provides a natural pathway to scaling this approach to high-fidelity radiation hydrodynamics simulations on HPC clusters, transforming the human-in-the-loop design bottleneck into an automated, adversarially robust optimization cycle.

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